



Use of Cystatin C in Tandem with Creatinine to Improve GFR Estimation and Clinical Decision Making

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Abstract

Recognizing its clinical utility, the 2024 Kidney Disease Improving Global Outcomes (KDIGO) guideline recommends using equations that integrate serum creatinine and cystatin C for CKD diagnosis and staging. Cystatin C–inclusive equations generally outperform creatinine-based equations in bias and accuracy across diverse populations, including children and patients with comorbidities. Moreover, cystatin C–inclusive equations consistently demonstrate greater prognostic value than creatinine-only equations for clinical outcomes such as kidney failure with replacement therapy, cardiovascular disease, and mortality, particularly in individuals with large differences between creatinine-based and cystatin C–based eGFR. Among hospitalized patients and those undergoing significant changes in muscle mass, such as during weight loss or critical illness, cystatin C may improve GFR estimation and facilitate earlier detection of AKI. In clinical practice, cystatin C informs CKD staging, risk stratification, and medication dosing, particularly in scenarios where creatinine-based equations are less reliable. Although the integration of cystatin C into risk equations has shown mixed results, its inclusion enhances calibration in populations with frailty or large eGFR differences. Emerging evidence also supports the use of cystatin C in cancer care and kidney transplantation. Despite its promise, widespread clinical adoption of cystatin C faces several barriers, including in-house availability. To surmount these barriers, collaboration between nephrology and laboratory services is needed. As evidence grows, cystatin C–based approaches have the potential to transform CKD detection and management. This study synthesizes recent evidence on cystatin C's role in GFR estimation, prognostic utility, and clinical applications, highlighting emerging areas and strategies for its implementation in health care systems.

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Introduction

Compelling evidence has demonstrated cystatin C's ability to improve the accuracy of GFR estimation and prognostication of long-term clinical outcomes beyond serum creatinine. On this basis, the 2024 Kidney Disease Improving Global Outcomes (KDIGO) guideline on CKD recommended using GFR-estimating equations that combined creatinine and cystatin C.¹ In this narrative review, we summarize recent studies on cystatin C as a GFR marker and prognostic indicator, illustrate how cystatin C testing may inform clinical decision making, review emerging areas of its potential use, and offer a roadmap to facilitate its implementation within health care systems.

Use of Cystatin C in GFR-Estimating Equations

Cystatin C's potential as a filtration biomarker was first recognized in the 1990s.² Unlike creatinine, cystatin C is not derived from muscle catabolism and does not undergo tubular secretion (Figure 1).^{3–6} Consequently, compared with creatinine, cystatin C more strongly correlates

with measured GFR (mGFR; r^2 of approximately 0.91).⁷ However, neither is a perfect GFR biomarker because various non-GFR factors differentially affect them, as delineated in Figure 2.^{8–11}

GFR-Estimating Equations in Adult Populations

To account for interindividual differences, several equations adjust for age and sex when approximating mGFR with creatinine and/or cystatin C. Supplemental Table 1 summarizes the clinically predominant GFR-estimating equations: (1) CKD Epidemiology Collaboration (CKD-EPI)^{12,13} and (2) European Kidney Function Consortium (EKFC) equations.¹⁴ CKD-EPI equations include coefficients for creatinine, cystatin C, or both and were developed in more diverse populations than the EKFC equations.^{12,13} In contrast, the EKFC equations standardize creatinine and cystatin C against the Q factor, which represents the median biomarker value within age and sex strata in healthy populations, and, rather than a single equation with both biomarkers, the EKFC combined creatinine and cystatin C–based eGFR (eGFRcr-cys)

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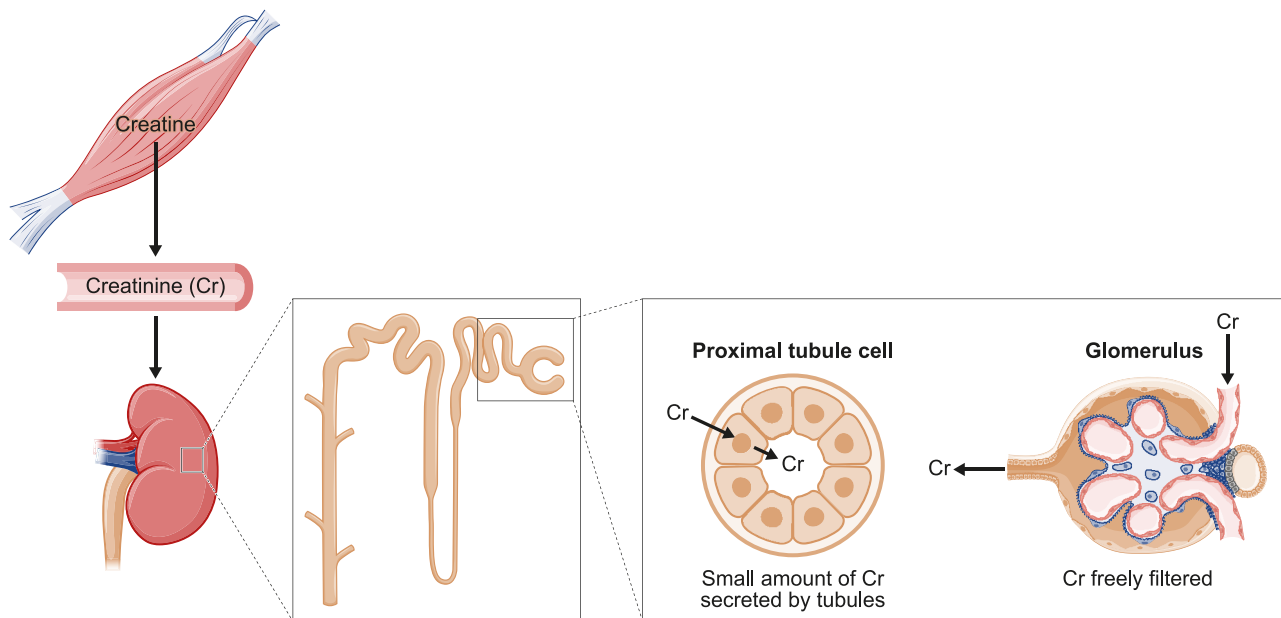
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A Creatinine generation and renal handling



B Cystatin C generation and kidney handling

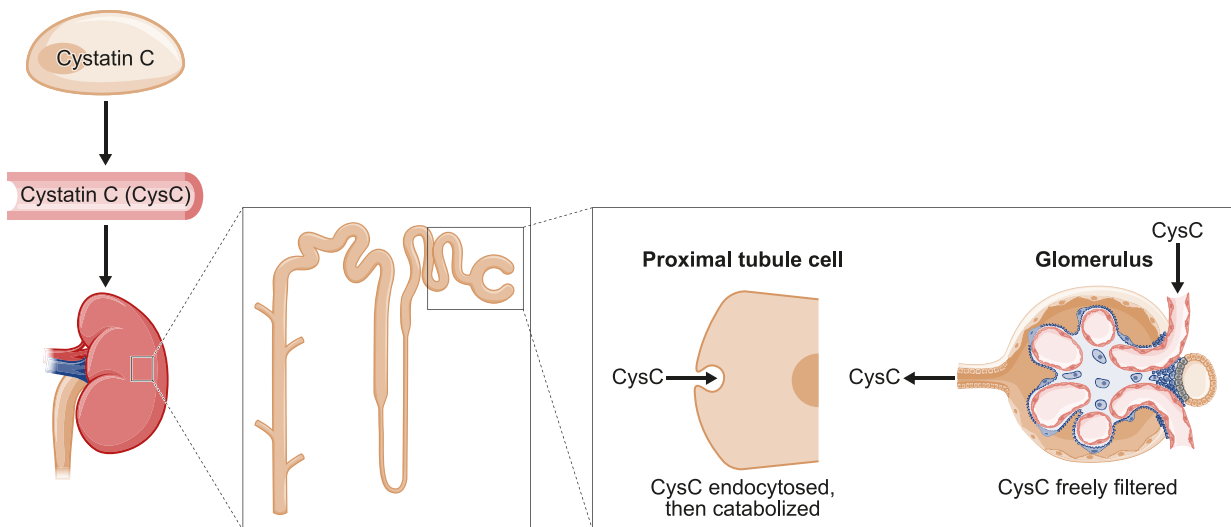


Figure 1. Creatinine and cystatin C physiology.^{3–6} (A) Creatinine generation and renal handling: Creatinine is a by-product of muscle catabolism. It is freely filtered within the glomerulus and undergoes secretion at the proximal tubules. (B) Renal cystatin C handling: Cystatin C is generated by all nucleated cells at a constant rate. It is freely filtered within the glomerulus and completely endocytosed by proximal tubules. Cr, creatinine; CysC, cystatin C.

represents the mean creatinine-based eGFR (eGFR_{cr}) and cystatin C-based eGFR (eGFR_{cys}) estimates.¹⁴ In addition, EKFC equations were developed in European White study populations and meant to be used throughout the lifespan.

Equation performances are evaluated by the degree to which they overestimate or underestimate mGFR (*i.e.*, bias in ml/min per 1.73 m²) and by the proportion of estimates within 30% of mGFR (*i.e.*, accuracy). Among the CKD-EPI equations, the eGFR_{cys} equation was the least biased (bias <1), whereas the 2021 eGFR_{cr-cys} was the most accurate (approximately 90%).¹³ Among EKFC equations,

eGFR_{cr-cys} was both the least biased (bias <1) and most accurate (84%–92%) across racial groups in Europe, the United States, and Africa.¹⁴ In general, the equations using both biomarkers compared with using creatinine alone have generally displayed more consistent performance across age, sex, and race subgroups and among those with comorbidities such as heart failure, liver disease, and cancer.^{15,16} Even among a subset of CKD-EPI participants in whom eGFR_{cr} and eGFR_{cys} differed substantially, eGFR_{cr-cys} yielded the least biased and most accurate estimates overall and across sex, age, and body

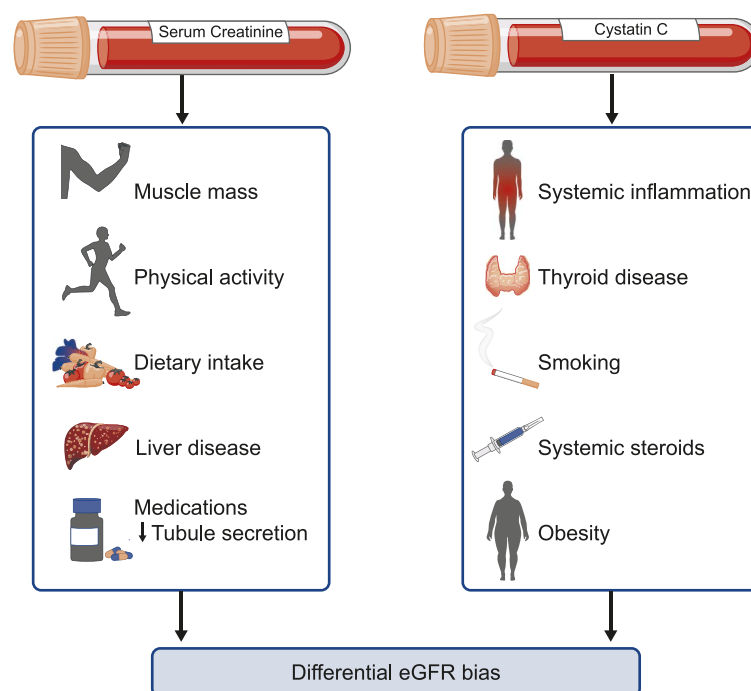


Figure 2. Non-GFR factors that influence exogenous filtration biomarkers.^{8–10}

mass index (BMI) strata.¹⁷ Similar findings were observed in a study of 4707 adults with mGFR in the Stockholm CREatinine Measurements cohort, which showed that eGFRcr-cys performed the best across the wide range of BMIs, including BMIs <20 or >30 kg/m².¹¹ The totality of these observations highlight the advantage of using cystatin C along with creatinine and support the KDIGO guideline recommendation to use both biomarkers to estimate GFR for CKD diagnosis and staging.¹

However, a recent study suggests that all the current GFR-estimating equations may not adequately capture declines in GFR. In a study population comprising 895 predominantly White English adults with CKD¹⁸ followed over 3 years, mGFR fell from a baseline of 48 to 43 ml/min per 1.73 m²; all GFR-estimating equations underestimated mGFR change, particularly among persons who had progressed. Although CKD-EPI and EKFC eGFRcr-cys equations best approximated mGFR changes, the sensitivity for detecting CKD progression was <54%.¹⁸ Although these findings support the use of eGFRcr-cys equations to monitor kidney function, further studies are needed to determine clinical contexts wherein these equations may not adequately detect disease progression, especially within understudied populations.

GFR-Estimating Equations in Pediatric and Young Adult Populations

In contrast to adult populations, the optimal equation in the pediatric and young adult populations remains unclear as external validation remains limited for the two primary sets of equations (*i.e.*, CKD in Children Under 25 [CKiD-U25]¹⁹ and EKFC equations)¹⁴ (Supplemental Table 2). The CKiD-U25 equations using either biomarker were developed in a racially diverse population of children and young

adults with CKD,²⁰ whereas the EKFC equations were derived from European White populations, most of whom had well-preserved kidney function.^{2,14,21,22}

The CKiD-U25 eGFRcr equation had minimal bias in both the CKiD-U25 and EKFC study populations; similarly, the EKFC eGFRcr equation was minimally biased within European-based cohorts.^{23,24} Conversely, the bias for both CKiD-U25 and EKFC eGFRcys equations varied more substantially depending on the region, health status, and age of the study population.^{22–25} To our knowledge, the EKFC eGFRcr-cys equation has not been validated in a pediatric population, and none of the EKFC equations have been evaluated in racially diverse pediatric populations. Therefore, the KDIGO guideline¹ does not indicate a preferred equation and notes that eGFRcys equations may be more accurate among children with neurological disabilities, muscle wasting, metabolic disorders, or very low-protein diets. Additional validation studies across varied populations are needed to identify if there are ideal equations for diagnosing CKD among persons at-risk versus for monitoring CKD progression.

Implications of Incorporating Cystatin C in GFR Estimations on Drug Dosing

The US Food and Drug Administration (FDA) guides pharmaceutical manufacturers to conduct pharmacokinetic studies of drugs when over 30% of either the systemic drug or an unchanged metabolite is eliminated in the urine.²⁶ Historically, most drug labels recommend dosing based on creatinine clearance; however, the FDA recently updated their guidance to suggest using contemporary eGFR equations adjusted to an individual patient's BSA, expressed in ml/min, or the creatinine clearance equation.²⁶ Most often, drug dose adjustment is required

when the GFR is <50 ml/min, and most of the contemporary estimating equations perform similarly at this GFR range. The National Kidney Foundation further endorsed the transition from creatinine clearance to eGFR equations, including the use of BSA-adjusted eGFRcr-cys particularly among obese individuals.²⁷ In contrast, the European Medicines Agency recommends performing mGFR, where feasible. If mGFR is not available, eGFRcr may be used, but all GFR estimates must be BSA-adjusted and presented as absolute values.²⁸

Table 1 summarizes recent evidence regarding cystatin C's utility in drug dosing, particularly for antimicrobial agents, direct oral anticoagulants (DOACs), and chemotherapy. eGFRcr-cys generally improves dosing and therapeutic exposure in patient populations who have a high prevalence of non-GFR determinants of creatinine. However, study results vary because of heterogeneity in patient populations and study designs, and most studies lack mGFR as a reference standard, limiting the interpretation of the results. Some antimicrobials, such as cefepime or teicoplanin, do not demonstrate superiority of eGFRcr-cys to eGFRcr; however, studies of vancomycin demonstrate benefit.^{29–41} This underscores the importance of conducting prospective pharmacokinetic studies where measured drug concentrations are correlated with mGFR and BSA-adjusted eGFR.

Among the DOACs, eGFRcr-cys does not appear to offer superiority to eGFRcr-based dosing, even for dabigatran and rivaroxaban, which undergo significant elimination by the kidneys (Table 1). In a study by Villain *et al.*, dose status (appropriately dosed, underdosed, or overdosed) was not associated with mortality and thromboembolism, regardless of which eGFR equation was used.⁴² Therefore, use of cystatin C-based approaches to dose DOACs should be reserved for borderline dosing decisions; alternatively, the use of AntiXa concentrations to guide dosing could be considered.

The American Society of Onco-Nephrology recommends using eGFRcr-cys to inform decision making related to anticancer therapy eligibility and dose adjustment and avoiding the Cockcroft–Gault formula.⁴³ They recommend cautious use of BSA adjustment for carboplatin dosing, while standardized (*i.e.*, BSA-indexed) GFR estimates are deemed appropriate for dose adjustments of other anticancer drugs.⁴³

Utility of Cystatin C in Prognostication and Risk Stratification

Cystatin C's Prognostic Performance

Cystatin C's prognostic utility consistently exceeds that of creatinine.^{44–46} A meta-analysis of 820,000 persons with concurrent creatinine and cystatin C measurements compared associations of eGFRcr, eGFRcys, and eGFRcr-cys with ten clinical outcomes.⁴⁷ In general, lower eGFR was associated with higher risks of all-cause and cardiovascular mortality, cardiovascular events, kidney failure with replacement therapy (KFRT), AKI, and hospitalizations; however, the magnitudes of associations were stronger and more linear for eGFRcr-cys compared with eGFRcr, which had weaker, U-shaped associations with clinical

outcomes. Such prognostic differences between creatinine-based versus cystatin C-based equations have occurred across varied populations and clinical contexts, including adults aged ≥65 years, South Asian populations, those with cirrhosis, and recently hospitalized patients.^{48–52}

In addition, large differences between eGFRcr and eGFRcys (or eGFRcr-cys), defined as discordance of ≥15 ml/min per 1.73 m² or ≥30% (≥15% for eGFRcr and eGFRcr-cys differences), occur commonly and have been shown to correlate with adverse outcomes.^{46,53,54} Among >800,000 individuals, an eGFRcys at least 30% lower than eGFRcr⁴⁶ was associated with a higher risk of cardiovascular events, incident heart failure, KFRT, and all-cause mortality compared with those with concordant eGFRs. Conversely, those in whom eGFRcys or eGFRcr-cys was significantly higher than eGFRcr had lower risk of these outcomes compared to those with concordant eGFRs. These associations have held across different clinical contexts, including acute heart failure, major surgical procedures, post-hospital discharge, and AKI recovery.^{50,51,55,56} Furthermore, in studies with serial eGFRcr and eGFRcys assessments, individuals in whom eGFRcys became progressively lower than eGFRcr experienced a higher risk of incident heart failure and all-cause mortality compared with those with concordant eGFRs over time.^{54,57} These observations implicate that equations with cystatin C and large eGFR differences may identify high-risk individuals or those with declining health. These associations are not causal. Rather, they reflect improved estimation of true GFR and its associated risk with adverse outcomes and the associations of non-GFR factors that lower creatinine (*e.g.*, frailty and low physical activity) or increase cystatin C (*e.g.*, inflammation, smoking) with worse outcomes.

Potential Role of Cystatin C in Risk Stratification

Risk scores that include eGFRcr have been developed for predicting: (1) incident cardiovascular disease using the American Heart Association's Predicting Risk of cardiovascular disease EVENTS (PREVENT) equations⁵⁸; (2) mortality risk for persons with CKD using the mortality risk equation in patients with CKD (MREK) equation⁵⁹; (3) and KFRT in persons with CKD by the kidney failure risk equation (KFRE).⁶⁰ These equations aim to discriminate risk for the specific outcome across patients (*i.e.*, c-statistic) and accurately quantify the individual's risk during a specified time frame (*i.e.*, calibration).

To our knowledge, no publication has evaluated the utility of including cystatin C into the PREVENT equations. Therefore, studies are needed to determine whether inclusion of eGFRcys within PREVENT equations would also improve cardiovascular prediction. However, because these equations include many cardiovascular risk factors, eGFRcys is unlikely to meaningfully improve discrimination unless the population is enriched for patients with large eGFR differences.

The MREK equation quantifies the 5-year mortality risk for older adults with CKD, recognizing that for most, their competing risk of death far exceeds their risk of KFRT.⁵⁹ Among older adults in the validation study (mean age, 80 years; eGFRcr, 50 ml/min per 1.73 m²), adding eGFRcys to the MREK equation improved the c-statistic from

Table 1. Summary of studies investigating the use of cystatin C for drug dosing

Author and Year	Methods and Sample Size	Primary Results	Interpretation
Antimicrobials			
Barreto <i>et al.</i> , 2023 ²⁹	Prospective PK modeling study in 120 critically ill adults receiving cefepime	The 2012 CKD-EPI eGFRcr-cys expressed in ml/min was most closely associated with cefepime CL. The observed concentration versus population predicted cefepime concentrations resulted in an R^2 of 0.67, MAPE of 42.7%, and MPE of 19.9%. The observed concentration versus individual predicted concentrations resulted in an R^2 of 0.91, MAPE of 20.5%, and MPE of 9.4%	Demonstrates evidence for combined biomarker kidney function estimation in ICU populations with sepsis where accurate dosing is critical for clinical outcomes
Downes <i>et al.</i> , 2020 ³⁰	Prospective observational study of 80 critically ill children receiving vancomycin to compare eGFRcys and eGFRcr-cys with Schwartz, to estimate vancomycin CL	eGFRcys improved model fit more than the bedside Schwartz equation. Including the full age spectrum equation, an eGFRcr-cys, improved model most; female sex was also a significant covariate of CL in the model. Final model PK indices were CL=0.29 L/h per kilogram and Vd=0.48 L/kg	eGFRcr-cys improves prediction of vancomycin CL compared with bedside Schwartz in critically ill children
Frazer <i>et al.</i> , 2017 ³¹	Preintervention and postintervention quality improvement study in ICU patients with stable kidney function. Compared vancomycin dosing using eGFRcr-cys nomogram versus traditional creatinine clearance-based dosing. Evaluated steady-state trough achievement with adjustment for confounders (age, sex, APACHE III score, fluid balance, sepsis)	Goal trough achievement: 50% (67/135) in the cystatin C intervention arm versus 28% (74/264) in the control arm (OR, 2.53; 95% CI, 1.65 to 3.90; $P < 0.001$). After adjustment for confounders, OR improved to 2.79 (95% CI, 1.76 to 4.44; $P < 0.001$). Similar clinical outcomes between groups	eGFRcr-cys-based vancomycin dosing nearly doubles the likelihood of achieving therapeutic targets in ICU patients. Robust benefit maintained after adjustment for confounders
Ibe <i>et al.</i> , 2025 ³²	Retrospective cohort of 118 patients with simultaneous serum creatinine and cystatin C at vancomycin initiation	72 patients with eGFR discordance (serum creatinine-based $\geq 30\%$ higher than cystatin C-based) had significantly higher measured vancomycin AUC than predicted ($P < 0.001$). Multivariable analysis identified age, hospital stay ≥ 12 d, and eGFR discordance as predictors of AUC upward deviation	Highlights importance of identifying discordance in kidney function estimates using serum creatinine versus cystatin C in relation to drug accumulation. Prolonged hospitalization or ICU stay more likely to have higher AUC
Jonckheere <i>et al.</i> , 2016 ³³	PK modeling study using NONMEM in 20 ICU patients receiving cefepime. Analyzed 208 plasma and 51 urine samples over median 3-d follow-up. Evaluated serum creatinine, cystatin C, urea, uromodulin, measured creatinine clearance, and KIM-1 as covariates	Two-compartment model with renal/nonrenal CL components. Cockcroft–Gault was the best predictor, median absolute prediction error from 42.4% to 29.9%, between-person variability from 135% to 50%. Other serum creatinine and cystatin C-based formulas plus measured creatinine clearance performed similarly	In critically ill patients receiving cefepime, multiple GFR estimation methods (including cystatin C) perform similarly to Cockcroft–Gault
Kozono <i>et al.</i> , 2016 ³⁴	Compared teicoplanin concentration prediction using cystatin C (Hoek equation) versus serum creatinine (Cockcroft–Gault) in hospitalized patients. Population mean method for initial dose adjustment	Predicted creatinine clearance (Cockcroft–Gault): median 45.23 ml/min (IQR, 32.12–58.28). Predicted GFR (Hoek): median 45.23 ml/min (IQR, 35.40–53.79). Root mean-squared prediction error similar: 6.88 (IQR, 3.80–9.96) $\mu\text{g/ml}$ for serum creatinine versus 6.72 (IQR, 3.77–9.68) $\mu\text{g/ml}$ for cystatin C. No significant difference between methods	For teicoplanin dosing, cystatin C provides no advantage over standard creatinine-based approach
Kees <i>et al.</i> , 2016 ³⁵	Population PK study of meropenem continuous infusion in 32 surgical ICU patients comparing measured creatinine clearance, Cockcroft–Gault, Hoek formula, 1/serum creatinine, 1/cystatin C as covariates	All kidney function estimates improved model performance. Strongest association with CL: measured creatinine clearance (typical CL=11.3 L/h $\times [1+0.00932 \times (\text{creatinine clearance}-80)]$), followed by 1/cystatin C. Cockcroft–Gault was least predictive. Age, weight, and sex not significant covariates	For meropenem continuous infusion in ICU, measured creatinine clearance optimal but cystatin C provides promising alternative. Outperforms Cockcroft–Gault. Supports cystatin C as practical option for carbapenem dosing in critical care

Table 1. Continued

Author and Year	Methods and Sample Size	Primary Results	Interpretation
Matsubara et al., 2021 ³⁶	Prospective PK study of ampicillin/sulbactam in elderly patients with pneumonia	Ampicillin PK parameters: CL=6.5±4.0 L/h, Vd=19.3±0.2 L, Ke=0.4±0.2, and $t_{1/2}$ =2.7 ± 1.6 h eGFRcys most correlated (r =0.7374). CL of ampicillin=0.1937× eGFRcys-c-0.6726	Used drug concentrations to find best correlate of kidney function. eGFRcys performed well for elderly patients
Meesters et al., 2018 ³⁷	Prospective PK study of ciprofloxacin in 22 children with complicated UTIs	Fat-free mass and eGFRcr-cys were significant covariates for ciprofloxacin CL. CL ranged 0.16–0.43 L/h per kilogram; Vd 0.06–2.88 L/kg. 100% achieve target for <i>Enterobacteriaceae</i> , 53% for <i>Pseudomonas aeruginosa</i> , 3% for <i>Staphylococcus aureus</i>	First pediatric study demonstrating cystatin C utility for fluoroquinolone dosing. Body composition and kidney function both critical in children with wide age range
Petersson et al., 2016 ³⁸	Prospective study of 20 patients on piperacillin/tazobactam and 19 on meropenem. Measured antibiotic concentrations and compared eGFRcr, eGFRcys, and measured creatinine clearance	Subtherapeutic concentrations at mid and end of dosing interval, 27% and 61% for piperacillin and 6% and 48% for meropenem. Lower GFR estimates associated with higher concentrations, but ten-fold variation existed between patients with similar kidney function. eGFRcys and measured creatinine clearance showed similar correlation with drug half-life	Standard dosing results in subtherapeutic concentrations in significant proportion of ICU patients. Wide interpatient variability persists even with similar kidney function estimates. Supports need for therapeutic drug monitoring alongside kidney function assessment
Teaford et al., 2020 ³⁹	Retrospective study of overweight or obese patients receiving vancomycin comparing four different kidney function equations (Cockcroft–Gault, eGFRcr, eGFRcys, eGFRcr-cys) for attainment of target concentration	The optimal model to predict the initial vancomycin trough included both cystatin C and serum creatinine (R^2 =0.48) rather than either biomarker alone. This model predicted 79% (95% CI, 73% to 85%) of troughs at goal compared with the 62% observed in clinical practice (P < 0.001), representing 1.3-fold improvement	Obese or overweight patients have suboptimal attainment of target concentrations. Addition of cystatin C to kidney function estimation improves predicted target attainment
Wansing et al., 2025 ⁴⁰	Retrospective study in hematopoietic stem cell transplant patients receiving vancomycin to compare eight kidney function–estimating equations to vancomycin CL	Cystatin C–based equations showed significantly better accuracy (bias: 6.3–9.5 ml/min per 1.73 m ²) versus creatinine–based equations (bias: 24.0–62.8 ml/min per 1.73 m ²) for predicting renal vancomycin CL. Superiority most pronounced in patients with reduced muscle mass or receiving glucocorticoids	Critical for immunocompromised populations where muscle wasting is common. CKD-EPI and Hoek equations are more precise than Cockcroft–Gault
Yoo et al., 2025 ⁴¹	PK study of benzylpenicillin in 72 healthy volunteers with bolus and continuous infusion dosing and evaluated which kidney function marker (serum creatinine, cystatin C, or combined eGFR) best predicted CL	Two-compartment model with concentration-dependent CL. eGFRcys significantly enhanced model fit compared with eGFRcr for predicting CL. Higher CL noted at very low concentrations (3 ng/ml)	Demonstrates cystatin C advantage for penicillin G dosing in healthy volunteers. This may improve dosing for long-acting preparations used for treatment of syphilis and prevention of rheumatic fever
DOACs			
Calsolaro et al., 2022 ⁹⁷	Post hoc analysis of prospective cohort study in hospitalized older adults receiving DOACs	Adding cystatin C to eGFR calculations reallocated one third of patients to worse kidney function category. CKD-EPI demonstrated best prediction of bleeding events (AUROC, 0.71). However, bleeding prevalence increased only 2%–3% in reclassified patients without reaching statistical significance	In elderly populations, cystatin C reclassifies substantial proportion but clinical impact on bleeding risk modest. eGFRcr-cys was best predictor of bleeding events (AUROC, 0.71). May be most useful for borderline dosing decisions rather than routine use
Lin et al., 2021 ⁹⁸	Observational study of 511 patients with atrial fibrillation on DOACs comparing kidney function equations	Both Cockcroft–Gault creatinine clearance and eGFRcr-cys identified patients with higher-than-expected dabigatran or rivaroxaban concentrations. Neither equation associated with high apixaban concentrations. Among elderly patients (≥75 yr) using dabigatran, CKD-EPI may be associated with increased drug concentrations	Suggests cystatin C equations may identify high-risk patients for certain DOACs (dabigatran, rivaroxaban) but not others (apixaban). Age-specific considerations are important. Drug-specific differences complicate uniform recommendations

Table 1. Continued

Author and Year	Methods and Sample Size	Primary Results	Interpretation
Chemotherapy			
Barreto <i>et al.</i> , 2022 ⁹⁹	Prospective PK study of 80 patients with lymphoma receiving high-dose MTX to determine which kidney function estimate best correlated with MTX CL	Median CL 4.5–5.6 L/h. Liver function, weight, age, sex, concomitant chemotherapy, and number of previous MTX doses did not affect CL. The eGFRcys most closely correlated with MTX CL in both the entire cohort and after excluding outlier MTX CL values ($r=0.31$ and 0.51 , respectively)	Provides evidence for the use of cystatin C for dosing chemotherapy with significant renal CL
Bretagne <i>et al.</i> , 2017 ¹⁰⁰	Prospective study of 131 patients with cancer without kidney insufficiency; 24 received carboplatin AUC 5	In patients with lowest LBM percentage, creatinine clearance significantly overestimated GFR versus cystatin C–based GFRcr-cys ($P = 0.0004$). Creatinine clearance/GFRcr-cys ratio >1.23 predicted severe thrombocytopenia ($P = 0.006$). LBM correlated with serum creatinine ($r=0.30$, $P < 0.005$) but not cystatin C ($r=-0.07$, $P = 0.43$)	eGFRcr overestimation leads to carboplatin overdosing and severe hematologic toxicity. Ratio >1.23 is an actionable threshold for switching to cystatin C–based dosing
de Rouw <i>et al.</i> , 2024 ¹⁰¹	Population PK analysis clinical trial data of pemetrexed PK from 66 participants and different kidney function estimates	eGFRcr-cys improved modeling of pemetrexed CL, explaining 35% of the total interindividual variability in CL	Kidney function estimates using serum creatinine and cystatin C perform best for predicting pemetrexed CL

APACHE, acute physiology and chronic health evaluation; AUC, area under the curve; AUROC, area under receiver operator curve; CI, confidence interval; CKD-EPI, CKD Epidemiology Collaboration; CL, clearance; DOAC, direct oral anticoagulant; eGFRcr, creatinine-based eGFR; eGFRcr-cys, combine creatinine and cystatin C–based eGFR; eGFRcys, cystatin C–based eGFR; ICU, intensive care unit; IQR, interquartile range; KIM-1, kidney injury molecule-1; LBM, lean body mass; MAPE, mean absolute percentage error; MPE, mean percentage error; MTX, methotrexate; NONMEM, nonlinear mixed effects modeling; OR, odds ratio; PK, pharmacokinetic; R^2 , coefficient of determination; UTI, urinary tract infection; Vd, volume of distribution.

approximately 70% to approximately 73%. This demonstrates the potential prognostic value of eGFRcys relative to eGFRcr among older adults with CKD.

Addition of eGFRcys to the KFRE has yielded mixed results. Among persons with moderate-to-advanced CKD, the KFRE using eGFRcr had a c-statistic of approximately 0.95 and did not differ across eGFR equations with or without cystatin C.⁶¹ However, in frail individuals, particularly those with substantial eGFRcr and eGFRcys differences, use of eGFRcys improved KFRE calibration.⁶² Moreover, the Z6 KFRE, which included cystatin C along with creatinine, urea, hemoglobin, and urine albumin-creatinine ratio, substantially outperformed KFRE (c-statistic 0.91 versus 0.86).⁶³ When externally validated, Z6 performed modestly better than the KFRE, with increases in c-statistic of 0.01–0.03.

At a population-level, these results suggest that the potential value of incorporating eGFRcys depends on the prevalence of large differences from eGFRcr and the relative prognostic superiority of cystatin C, which is greatest for outcomes of mortality and heart failure.

Emerging Areas of Investigation

Implications for Cystatin C Testing in Hospitalized Patients

Evidence of cystatin C's utility in the inpatient setting has lagged despite high proportions of hospitalized patients with large differences between eGFRcr and eGFRcys.

Among approximately 39,000 hospitalized patients across two inpatient cohorts who underwent clinical cystatin C testing, 35% had an eGFRcys that was at least 30% lower than eGFRcr.⁴⁶ However, patients with cystatin C testing were clinically selected and represented $<5\%$ of the overall inpatient populations; thus, the estimated prevalence of large eGFR differences was likely inflated. Nonetheless, these differences have important implications on how clinicians should monitor kidney function and detect AKI among inpatients.

Unfortunately, evidence delineating which existing eGFR equation best approximates mGFR in this setting remains limited. The largest study to date comprised >3000 hospitalized Spanish adults with preserved kidney function who underwent mGFR and showed that eGFRcys was less biased and more accurate than the 2009 CKD-EPI eGFRcr and eGFRcr-cys equations, particularly among older patients (≥ 70 years) or those with cirrhosis or malnutrition.⁶⁴ The performance difference was largely explained by muscle mass's influence on eGFRcr, with minimal effect from systemic inflammation, diabetes, or thyroid disease. These findings were further supported by an in-depth examination of 32 mechanically ventilated, critically ill patients, 27 of whom underwent mGFR assessment.⁶⁵ During their intensive care unit (ICU) stay, eGFRcr became progressively higher than eGFRcys, with the difference widening from an average of 2 ml/min per 1.73 m^2 at admission to 33 ml/min per 1.73 m^2 at ICU discharge; this was partially explained by decreases in

muscle mass. However, both equations substantially overestimated mGFR at discharge (bias: eGFRcys 22, eGFRcr 59),⁶⁵ suggesting either the preponderance of additional non-GFR factors influencing creatinine or cystatin C or potential effect of changes in fluid compartments on mGFR's accuracy in this setting.^{66–68}

Meta-analyses of cystatin C's predictive performance for AKI defined by creatinine elevations have reported pooled sensitivities of 78%–86% and specificities of 79%–87%, with c-statistics of 0.86–0.89.^{69,70} Among 85 individuals at a high risk of AKI, cystatin C increased by $\geq 50\%$ on average 1.5 days before creatinine, suggesting that cystatin C may have a role in earlier AKI detection.⁷¹ However, existing studies have varied substantially with regard to clinical settings and AKI definitions, and all relied on creatinine-based AKI definitions.^{69,70} In sum, future investigations warrant larger study populations and attention to clinically compromised patients such as those in the ICU or with decompensated heart failure to clarify the role of cystatin C in kidney function monitoring and AKI diagnosis during hospitalizations.

Cystatin C's Potential Value in Cirrhosis

Creatinine remains the filtration marker used in the Model for End-stage Liver Disease (MELD) score.⁷² This likely underestimates a patient's liver disease severity and affects their relative priority listing for liver transplantation. However, a cystatin C–based MELD score with cirrhosis did not improve prediction of 90-day mortality among 429 patients.⁷³

Most of the evidence for cystatin C–inclusive eGFR equations in patients with cirrhosis predate the EKFC and 2021 CKD-EPI equations. To our knowledge, no study has evaluated the EKFC equations in cirrhosis, and only one small study has compared the 2021 CKD-EPI equations with mGFR. Among 137 patients with cirrhosis and a median mGFR of 77 ml/min per 1.73 m², all GFR-estimating equations overestimated mGFR (mean bias of 4.5, 15.0, and 9.7 for eGFRcr, eGFRcys, and eGFRcr-cys, respectively).⁵² Among the 44 patients with a mGFR ≤ 60 ml/min per 1.73 m², eGFRcys was the least biased, with a mean bias of 3.7 versus 14.6 for eGFRcr and 7.6 for eGFRcr-cys. Given the importance of accurate GFR estimation in detecting kidney disease and liver transplant decision making, larger studies evaluating the contemporary GFR-estimating equations in cirrhosis are urgently needed.

Cystatin C Use in Patients with Cancer

Kidney function has important implications for treatment eligibility and drug dosing among patients with cancer. The American Society of Onco-Nephrology recently endorsed using eGFRcr-cys⁴³ based on its superior performance among persons with cancer, including multiracial populations.^{74,75} Among 1145 individuals with cancer and mGFR, the 2021 CKD-EPI eGFRcr-cys equation performed the best among the CKD-EPI, EKFC, and Cockcroft–Gault equations, with a median bias of 3% and 86% accuracy; this held across hematologic and solid malignancies.⁷⁵

Differences in equation performances may partially result from heterogeneity in body composition. Among 465

patients with cancer, of whom one third had sarcopenia on computed tomography imaging, eGFRcr and eGFRcys performed poorly in sarcopenic patients while eGFRcr-cys performed well even in this subgroup.⁷⁶ However, the best performing eGFR equation incorporated creatinine and cystatin C along with $\beta 2$ -microglobulin and β -trace protein, with 92% accuracy. This body of work supports the use of the 2021 CKD eGFRcr-cys equation among patients with cancer, but additional biomarkers may further enhance accuracy.

Cystatin C's Utility in Monitoring Kidney Function during Weight Loss

Bariatric surgery and glucagon-like protein-1 receptor agonists result in substantial fat and muscle loss, with the latter accounting for approximately 25% of weight loss.⁷⁷ This may interfere with creatinine and cystatin C. Among 105 patients undergoing bariatric surgery, preoperatively, all eGFR equations substantially underestimated mGFR when indexed to a BSA of 1.73 m² and had poor accuracy. The mean weight declined by 36 kg, and the mean mGFR decreased from 107 to 92 ml/min per 1.73 m² after surgery.⁷⁸ Postoperatively, eGFRcys and eGFRcr-cys remained suboptimal, but eGFRcr-cys had minimal bias (< 2 ml/min) and improved accuracy (82%–85%) in presurgery and postsurgery periods when BSA adjusted. A study of 27 patients who underwent bariatric surgery aligned with these findings.⁷⁹ These findings may reflect the poor performance of all eGFR equations above 60 ml/min per 1.72 m². Although similar findings would be anticipated in pharmacologically induced weight loss, to our knowledge, no study with mGFR has been published in this specific context.

Cystatin C Use in Kidney Transplantation

The 2017 KDIGO guideline on living kidney donor evaluation recommends a two-step kidney assessment.⁸⁰ First, eGFRcr serves as the initial screening test. Among individuals with eGFRcr ≥ 60 ml/min per 1.72 m², eGFRcr-cys, 24-hour urine creatinine clearance, or mGFR may be used for confirmation. Postdonation, GFR generally drops to approximately 70% of predonation levels⁸¹; however, no evidence exists regarding cystatin C's utility in monitoring kidney function among living donors.

In pretransplant recipient evaluation, eGFR aims to identify individuals whose GFR has reached < 20 ml/min per 1.73 m² to determine transplant listing eligibility. In 2022, the Organ Procurement and Transplantation Network began requiring race-neutral GFR-estimating equations, allowing use of the CKD-EPI eGFRcys or 2021 eGFRcr and eGFRcr-cys equations to determine whether a patient meets this GFR cutoff.^{82,83} This includes medical record reviews of Black patients to ascertain when their GFR fell below 20 ml/min per 1.73 m² using a race-neutral eGFR and adjust their transplant wait time^{83,84}; no evidence exists to inform which specific eGFR equation to use for this purpose.

Contemporary studies that evaluated cystatin C–inclusive eGFRs in kidney allograft function assessments yielded conflicting conclusions. Among 415 recipients who were on average 7 years post-transplant and had a mean

mGFR of 53 ml/min per 1.73 m², all CKD-EPI equations were minimally biased; eGFRcys had the lowest accuracy overall and across eGFR strata compared with eGFRcr and eGFRcr-cys (overall 74% versus 83% and 84%).⁸⁵ Because eGFRcr and eGFRcr-cys had similar performances, cystatin C's usefulness in post-transplant monitoring was unclear. A larger racially diverse study of recipients with a mean mGFR of 66 ml/min per 1.73 m² at 1 year post-transplant showed that all eGFR equations including EKFC eGFRcys had absolute biases of 13–16 ml/min per 1.73 m²; the 2021 CKD-EPI eGFRcr-cys had the highest accuracy at 78%, while CKD-EPI eGFRcys had the lowest (68%).⁸⁶ Studies are needed to investigate which post-transplant factors influence the performance of GFR-estimating equations and to optimize GFR estimation post-transplant.

Laboratory Considerations

Since certified reference materials (European Reference Materials-DA471/International Federation of Clinical Chemistry and Laboratory Medicine) for cystatin C were released in 2010,⁸⁷ all manufacturers with FDA-approved assays in the United States had at least one methodology with established traceability to the reference material by 2018. Furthermore, there has been marked improvement in both assay harmonization and accuracy compared with 2014 (Figure 3),⁸⁸ with the 2025 College of American Pathologists (CAP) proficiency testing data showing improved coefficients of variation (CVs) from 2014 to 2025, reflecting improved between- and within-method agreement.^{89,90}

However, differences in performance between cystatin C methods persist.⁹¹ Currently, cystatin C methods are available as homogeneous systems, wherein reagents are

designed to work on an instrument produced by the same manufacturer, or heterogeneous systems where reagents made by a third party are designed for use on instruments from different manufacturers (*i.e.*, open-channel assays). The 2025 CAP proficiency testing surveys showed mixed performances of both system types.^{89,90} Among homogeneous system reagents, Roche and Siemens consistently had CVs ≤5% for all samples, while the Binding Site Optilite had variable performance. Among heterogeneous system reagents, the Diazyme method was the only method with CVs consistently >5% for all samples, while Gentian reagents had CVs ≤5% for all but one sample, which was run on the Beckman AU/DxC AU Series instruments. Consequently, clinical laboratories should closely monitor their specific method's performance in proficiency testing surveys, and if unsatisfactory, apply pressure to their assay manufacturer to improve assay performance.

Implementation Strategies

The number of clinical laboratories that reported at least one result to CAP cystatin C surveys has dramatically risen (Figure 3).^{89,92–94} However, the assay still remains a send-out or reference laboratory test for most clinical laboratories.⁹⁵ This hinders clinical use because eGFRcr, a standard in-house test, would be available for decision making well before a send-out eGFRcys, and many health record systems are unable to autocalculate eGFRcr-cys from outside laboratory results.

Figure 4 outlines the steps to deploy in-house cystatin C testing. Laboratories must consider start-up and maintenance costs for cystatin C testing and whether the number of ordered tests would support these costs. Nephrology

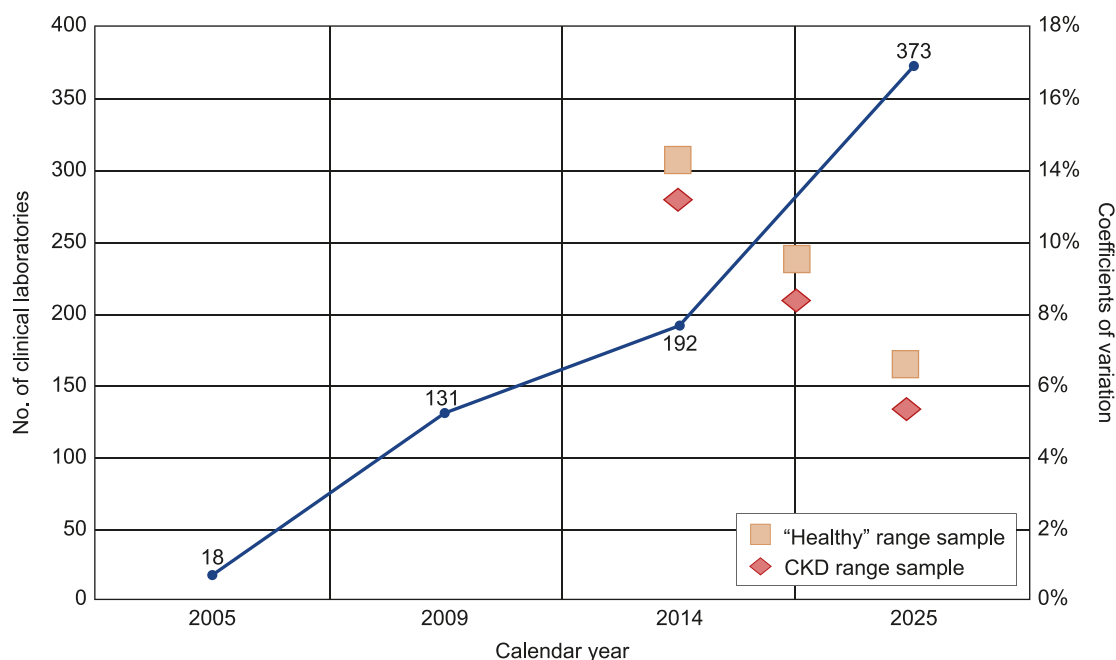


Figure 3. Number of clinical laboratories reporting results and coefficients of variation from the CAP cystatin C proficiency testing survey data, 2014–2025. Data presented were obtained from the CAP proficiency testing survey that is conducted semiannually to clinical laboratories within the United States and internationally. CAP, College of American Pathologists.

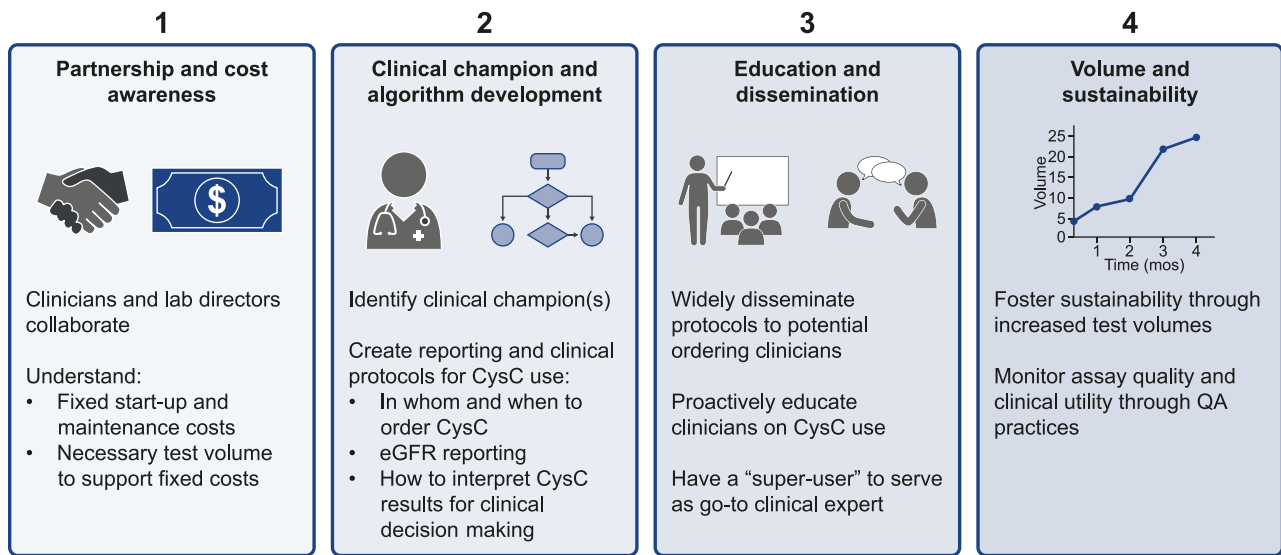


Figure 4. Roadmap to implementation of cystatin C in clinical practice. QA, quality assurance.

and clinical laboratory professionals must collaborate to develop protocols that outline in whom and when cystatin C should be ordered and how cystatin C–inclusive eGFRs should be reported and interpreted alongside eGFRcr.⁹⁶ Implementation requires a champion to lead these changes, disseminate knowledge, and serve as an expert clinical resource for ordering clinicians⁹⁶; this investment fosters sustainability by driving cystatin C test volumes and appropriate use.

Conclusion

Cystatin C has been established as a vital biomarker in clinical practice, improving the accuracy of GFR estimation and prognostication across diverse populations and clinical settings. Its integration into GFR-estimating equations alongside creatinine optimizes CKD detection, staging, and risk stratification and informs important medication dosing and clinical decision making. Emerging evidence hints at its potential utility in a variety of clinical contexts. Barriers to widespread implementation remain, including assay availability and clinical integration. Addressing these challenges to cystatin C use can substantially improve kidney care and patient outcomes.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at <https://links.lww.com/JASN/F930>.

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Supplemental Material

This article contains supplemental material online, published as provided by the authors, at <https://links.lww.com/JASN/F931>.

Supplemental Table 1. Summary of studies on the development and external validation of GFR-estimating equations.

Supplemental Table 2. General characteristics and applicability of eGFR equations in pediatric populations.

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